Scaling up:

Deep Reinforcement Learning

Ido Greenberg

2026

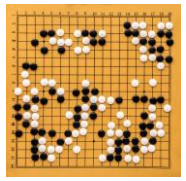
Tabular RL: Overview

	Environment	Known model: **Planning**	Unknown model		
			Model-based	**Model-free**	
				Value-based	Policy-based
1960s-1990s	**Tabular**	*Dynamic Programming*, *Value Iteration*, *Policy Iteration*	*Dyna*	*Q-learning*, *SARSA*, *Temporal Differences*	*Policy search*

Tabular RL: Limitation

- Tabular RL:
  - Consider each state $s \in S$ **individually**
  - Propagate values across states
  - **Complexity** $\geq \boldsymbol{O}(|\boldsymbol{S}|)$

- What if $|S| = 10^{80}$?



- What if $|S| = \infty$?



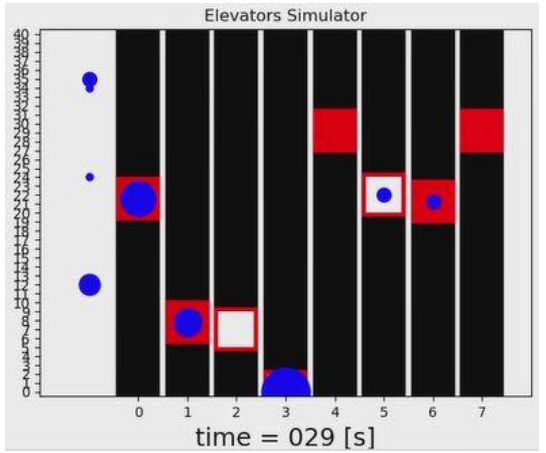
And if the problem *is* small?

- Example: optimize 2 elevators for 10 floors
- Actions: next elevator stops
- State space (simplified):
  - Location & dest. per elevator: $\sim n_{floors}^{2 \cdot n_{lifts}} = 10^4$
  - Location & dest. per user: $\sim n_{floors}^{2 \cdot n_{users}} \approx 10^6$

$$|S| \approx 10^{10}$$



בהשראת אלחנן אילת
1932-2025



Deep RL

- Function approximation: generalize over states
  - Value-based: $V_\theta(s)$ or $Q_\theta(s, a)$
  - Policy-based: $\pi_\theta(a|s)$

RL: Algorithm Overview

Environment	Known model: **Planning**	Unknown model		
		Model-based: Learn model, then plan	**Model-free**	
			Value-based	Policy-based
Tabular	*Dynamic Programming*, *Value Iteration*, *Policy Iteration* 	*Dyna*	*Q-learning*, *SARSA*, *Temporal Differences*	*Policy search*
Deep RL				

RL: Algorithm Overview

Environment	Known model: **Planning**
Tabular	*Dynamic Programming*, *Value Iteration*, *Policy Iteration* 
Deep RL	Approx planning *MCTS* 

<u>Approximate Planning</u>

(separate talk)

- Propagate rewards over states
- Deep RL: can't iterate *all* states
  - Learn *Policy*: which actions to sample
  - Learn *Value*: choose over sampled actions

RL: Algorithm Overview

Q-learning

$$Q(s,a) \mathrel{+}= \alpha \left(r + \max_{a'} Q(s',a') - Q(s,a) \right)$$

Curr value — r + next value — Curr value

- Propagate **next** → **current**

DQN

$$\text{argmin}_\theta \, E \left(r + \max_{a'} Q_\theta(s',a') - Q_\theta(s,a) \right)^2$$

Detach grads — Keep grads

Unknown model	
Model-free	
Value-based	Policy-based
Q-learning, *SARSA*, *Temporal Differences*	*Policy search*
DQN, *QR-DQN*	

Value-based

- “How good is this state/action?”

Value-based

- “How good is this state/action?” – hard to learn value

4.2. Unsuccessful Attempts

In the early stages of developing DeepSeek-R1, we also encountered failures and setbacks along the way. We share our failure experiences here to provide insights, but this does not imply that these approaches are incapable of developing effective reasoning models.

Monte Carlo Tree Search (MCTS) Inspired by AlphaGo (Silver et al., 2017b) and AlphaZero (Silver et al., 2017a), we explored using Monte Carlo Tree Search (MCTS) to enhance test-time compute scalability. This approach involves breaking answers into smaller parts to allow the model to explore the solution space systematically. To facilitate this, we prompt the model to generate multiple tags that correspond to specific reasoning steps necessary for the search. For training, we first use collected prompts to find answers via MCTS guided by a pre-trained value model. Subsequently, we use the resulting question-answer pairs to train both the actor model and the value model, iteratively refining the process.

However, this approach encounters several challenges when scaling up the training. First, unlike chess, where the search space is relatively well-defined, token generation presents an exponentially larger search space. To address this, we set a maximum extension limit for each node, but this can lead to the model getting stuck in local optima. Second, the value model directly influences the quality of generation since it guides each step of the search process. Training a fine-grained value model is inherently difficult, which makes it challenging for the model to iteratively improve. While AlphaGo’s core success relied on training a value model to progressively enhance its performance, this principle proves difficult to replicate in our setup due to the complexities of token generation.

In conclusion, while MCTS can improve performance during inference when paired with a pre-trained value model, iteratively boosting model performance through self-search remains a significant challenge.

DeepSeek-R1, *Nature*, 2025

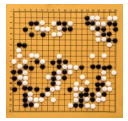


Value-based vs. Policy-based

- “How good is this state/action?”
  – hard to learn value
- “Which action is better?”
  – often easier

- Even when model is known

RL: Algorithm Overview

Environment	Known model: **Planning**	Unknown model		
		Model-based: Learn model, then plan	**Model-free**	
			Value-based	Policy-based
Tabular	*Dynamic Programming*, *Value Iteration*, *Policy Iteration* 	*Dyna*	*Q-learning*, *SARSA*, *Temporal Differences*	*Policy search*
Deep RL	Approx planning *MCTS* 	**World models** *MuZero*, *Dreamer*	*DQN*, *QR-DQN*	**Policy gradient / Actor-critic** *REINFORCE*, *PPO*  

Deep RL

- Policy-based: $\pi_\theta(a|s)$

- Representation:
  - **Discrete** actions: $\pi_\theta(\cdot\,|s) =$ vector with softmax
  - **Continuous** actions: parametric, e.g. $\pi_\theta(\cdot\,|s) = N\big(\mu_\theta(s), \sigma_\theta(s)\big)$

Policy Gradient (PG)

- Goal:

$$\max_{\theta} E_{\pi_\theta}[R]$$

- Method:
  - Play $\pi_\theta(a|s)$

Play π_θ for N episodes

Policy Gradient (PG)

- Goal:

$$\max_{\theta} E_{\pi_\theta}[R]$$

- Method:
  - Play $\pi_\theta(a|s)$
  - Good/bad outcome
  - Increase/decrease $\pi_\theta(a|s)$

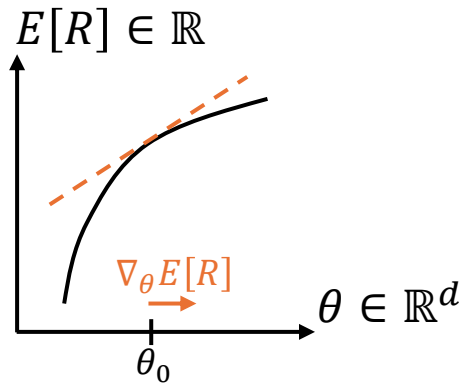


Policy Gradient (PG)

- Goal:

$$\max_{\theta} E_{\pi_\theta}[R]$$

- Method: move θ with $\nabla_\theta E[R]$
  - Like *Gradient Descent*



Policy Gradient (PG)

- $\boldsymbol{\nabla_\theta E[R]}$ from data:

Policy Gradient (PG)

- $\boldsymbol{\nabla_\theta E[R]}$ from data:
  - Write as $\boldsymbol{E[Y] = \int P_Y(y) \cdot y}$
  - Law of large numbers: $E[Y] \approx \frac{1}{N}\sum_{i=1}^{N} Y_i$

Policy Gradient (PG)

- $\boldsymbol{\nabla_\theta E[R]}$ from data:
  - Write as $\boldsymbol{E[Y] = \int P_Y(y) \cdot y}$

$$\boldsymbol{\nabla_\theta E_{\tau\sim\pi_\theta}[R]} = \nabla_\theta \int_\tau P(\tau|\theta)R(\tau) \qquad \text{expectation def.}$$

$$= \int_\tau \nabla_\theta P(\tau|\theta)R(\tau) \qquad \text{switch } \nabla/\int \text{ order}$$

Policy Gradient (PG)

- $\boldsymbol{\nabla_\theta E[R]}$ from data:
  - Write as $\boldsymbol{E[Y] = \int P_Y(y) \cdot y}$

$$\begin{aligned}
\boldsymbol{\nabla_\theta E_{\tau\sim\pi_\theta}[R]} &= \nabla_\theta \int_\tau P(\tau|\theta) R(\tau) && \text{expectation def.} \\
&= \int_\tau \nabla_\theta P(\tau|\theta) R(\tau) && \text{switch } \nabla/\int \text{ order} \\
&= \int_\tau P(\tau|\theta) \nabla_\theta \log P(\tau|\theta)\, R(\tau) && \nabla \log P = \frac{\nabla P}{P} \\
&= E_{\tau\sim\pi_\theta}[\nabla_\theta \log P(\tau|\theta)\, R(\tau)] && \text{expectation def.}
\end{aligned}$$

Policy Gradient (PG)

- $\boldsymbol{\nabla_\theta E[R]}$ from data:

$$\boldsymbol{\nabla_\theta E_{\tau\sim\pi_\theta}[R]} = E_{\tau\sim\pi_\theta}[\nabla_\theta \log P(\tau|\theta)\, R(\tau)]$$

$$\approx \frac{1}{N}\sum_{i=1}^{N} \nabla_\theta \log P(\tau_i|\theta)\, \underbrace{R(\tau_i)}$$

law of large numbers

Known from environment

Policy Gradient (PG)

- $\boldsymbol{\nabla_\theta E[R]}$ from data:

$$\boldsymbol{\nabla_\theta E_{\tau\sim\pi_\theta}[R]} = E_{\tau\sim\pi_\theta}[\nabla_\theta \log P(\tau|\theta)\, R(\tau)]$$

$$\approx \frac{1}{N}\sum_{i=1}^{N} \nabla_\theta \log P(\tau_i|\theta)\, R(\tau_i)$$ law of large numbers

$$= \frac{1}{N}\sum_{i=1}^{N}\sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_{i,t}|s_{i,t})\, R(\tau_i)$$ $P(\tau_i) \propto \Pi_t \pi_\theta(a_{i,t})$

Policy Gradient (PG)

- $\boldsymbol{\nabla_\theta E[R]}$ from data:

$$\boldsymbol{\nabla_\theta E_{\tau\sim\pi_\theta}[R]} = E_{\tau\sim\pi_\theta}[\nabla_\theta \log P(\tau|\theta)\, R(\tau)]$$

$$\approx \frac{1}{N}\sum_{i=1}^{N} \nabla_\theta \log P(\tau_i|\theta)\, R(\tau_i)$$ law of large numbers

$$= \frac{1}{N}\sum_{i=1}^{N}\sum_{t=0}^{T} \underbrace{\nabla_\theta \log \pi_\theta(a_{i,t}|s_{i,t})}\, R(\tau_i)$$ $P(\tau_i) \propto \Pi_t \pi_\theta(a_{i,t})$

Known: model output gradient

Policy Gradient (PG)

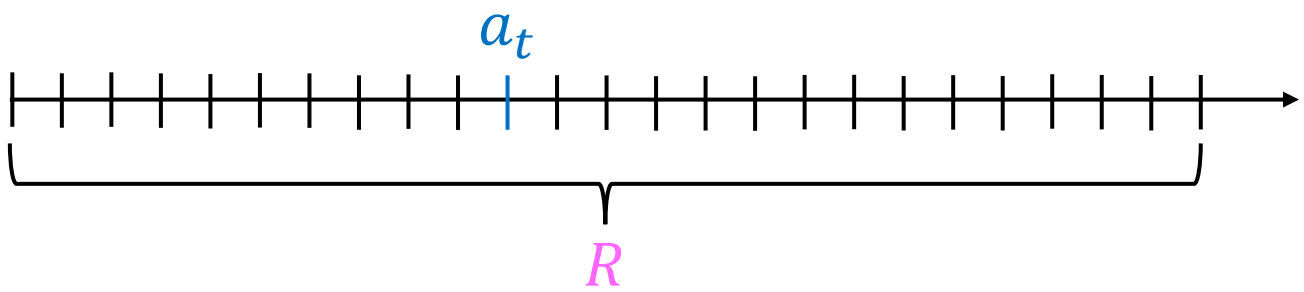
$$\nabla_\theta E_{\tau\sim\pi_\theta}[R] \approx \frac{1}{N}\sum_{i=1}^{N}\sum_{t=0}^{T} \nabla_\theta \log \pi_\theta\left(a_{i,t}|s_{i,t}\right) R(\tau_i)$$

High R $\Rightarrow$ change θ toward gradient $\Rightarrow$ increase $\pi_\theta(a)$ $\Rightarrow$ **encourage behavior**

- And vice versa

Challenge of Variance: Credit Assignment

- $\nabla_\theta \log \pi_\theta(a_t|s_t) \overbrace{\left(\sum_{k=0}^{T} \gamma^k r_k\right)}^{R(\tau)}$

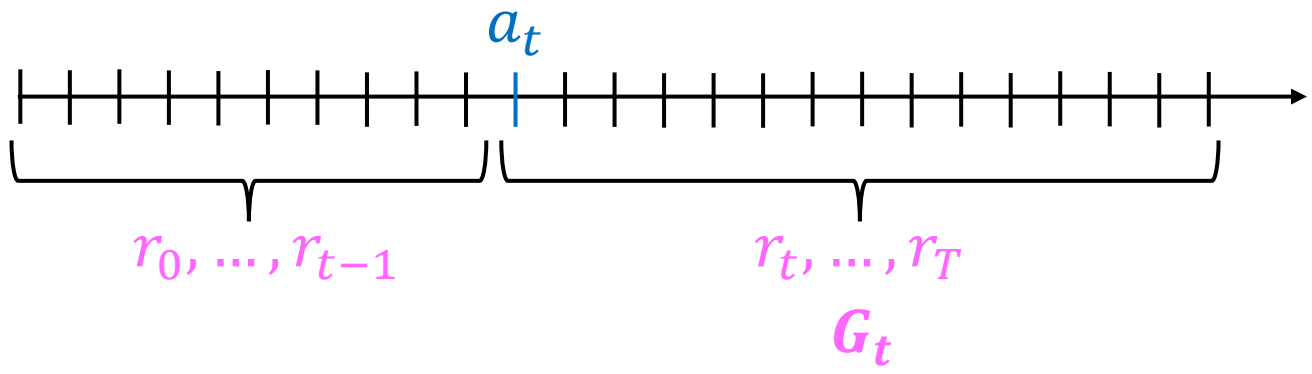


Challenge of Variance: Credit Assignment

- $\nabla_\theta \log \pi_\theta(a_t|s_t) \overbrace{\left(\sum_{k=0}^{T} \gamma^k r_k\right)}^{R(\tau)}$

- **Problem:** $r_0, \dots, r_{t-1}$ don't depend on a_t $\Rightarrow$ noise w/o information
  - Mathematically: $E\left[\nabla_\theta \log \pi_\theta(a_t|s_t) \left(\sum_{k=0}^{\boldsymbol{t-1}} \gamma^k r_k\right)\right] = 0$

- **Solution:** replace R with $G_t := \sum_{k=t}^{T} \gamma^{k-t} r_k$
  - Same expectation, lower variance



PG algorithm: REINFORCE

initialize θ

for epoch:

$\{\tau_i\}_{i=1}^{N} \leftarrow$ run N episodes with π_θ

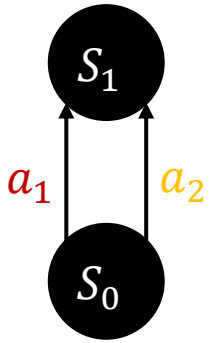
$G_{i,t} \leftarrow \sum_{k=t}^{T} \gamma^{k-t} r_{i,k}$ $(\forall i, t)$

$G \leftarrow \frac{1}{N} \sum_{i=1}^{N} \sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_{i,t}|s_{i,t}) \cdot G_{i,t}$

$\theta \leftarrow \theta + \alpha G$ (or another update rule like *Adam*)

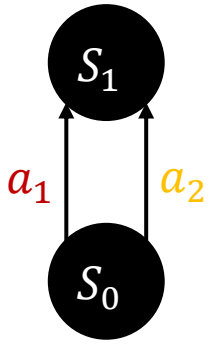
Challenge of Variance II

- Toy problem – 1 step, 2 actions:
  - $r(a_1) = 4$
  - $r(a_2) = 8$



Challenge of Variance II

- Toy problem – 1 step, 2 actions:
  - $r(a_1) = 4$
  - $r(a_2) = 8$



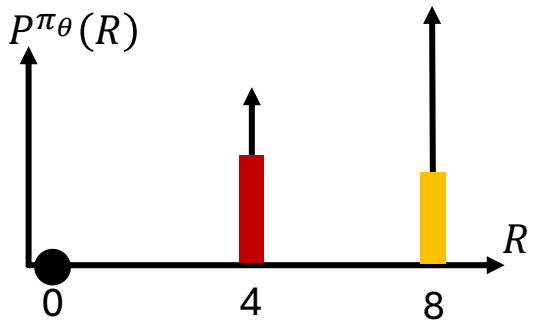
- $G \propto r(a) > 0$
  - *Both* a_1, a_2 are encouraged

Challenge of Variance II

- Toy problem – 1 step, 2 actions:
  - $r(a_1) = 4$
  - $r(a_2) = 8$

Unbiased: $E[G] = \nabla_\theta E[R]$

- $G \propto r(a) > 0$
  - *Both* a_1, a_2 are encouraged
  - a_2 grows faster – good asymptotically



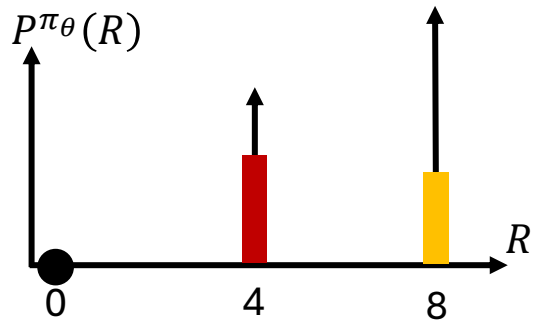
Challenge of Variance II

- Toy problem – 1 step, 2 actions:
  - $r(a_1) = 4$
  - $r(a_2) = 8$

Unbiased: $E[G] = \nabla_\theta E[R]$

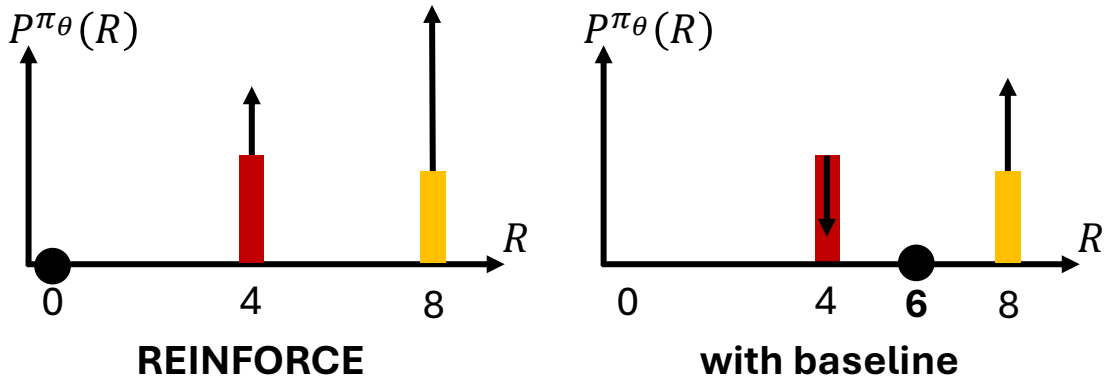
High $Var(G)$ over data

- $G \propto r(a) > 0$
  - *Both* a_1, a_2 are encouraged
  - a_2 grows faster – good asymptotically
  - **What if batch is $\{a_1, a_1, a_1, a_1\}$?**



Actor Critic (AC): PG + baseline

- Observation: mean$(R) \approx 6$
  - $G \leftarrow \frac{1}{N}\sum_{i=1}^{N} \nabla_\theta \log P(\tau_i|\theta)\,(R(\tau_i) - \boldsymbol{b})$ $\qquad (\boldsymbol{b} = V(S_0) = 6)$



Actor Critic (AC): PG + baseline

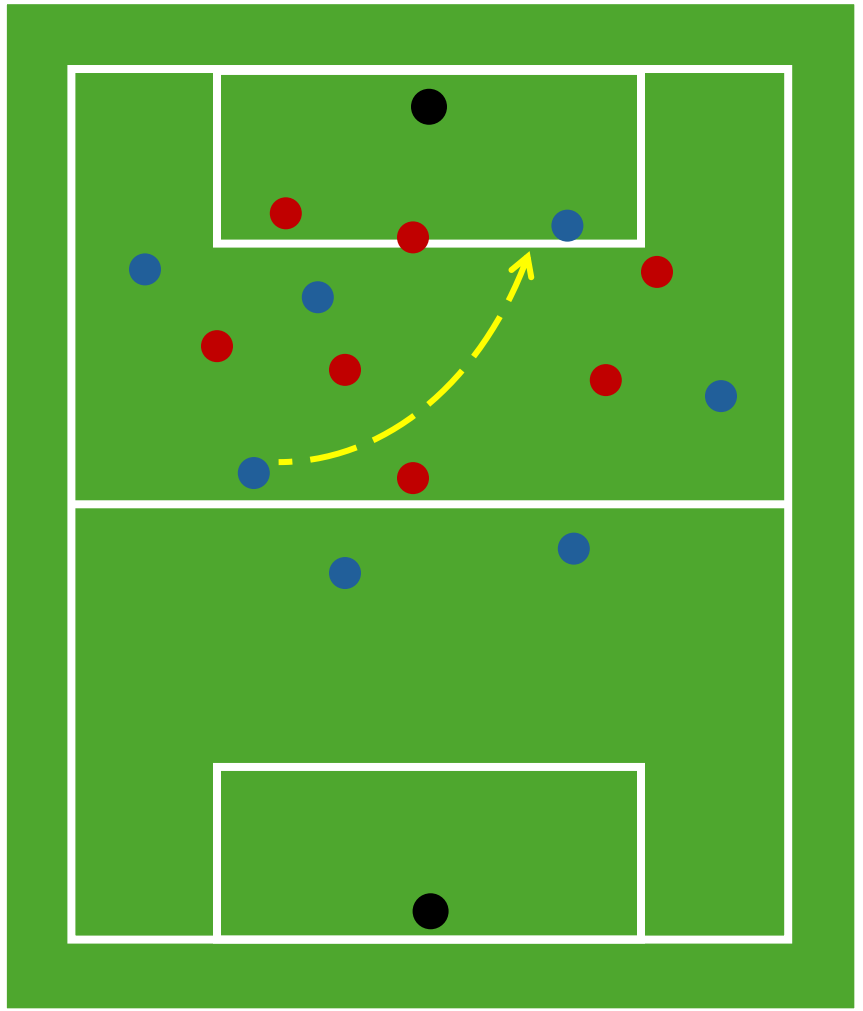
- Observation: mean$(R) \approx 6$
  - $G \leftarrow \frac{1}{N}\sum_{i=1}^{N} \nabla_\theta \log P(\tau_i|\theta)\,(R(\tau_i) - \boldsymbol{b})$ $\qquad (\boldsymbol{b} = V(S_0) = 6)$

- Theorem: $\boldsymbol{E[G] = \nabla_\theta E[R]}$
  - If b is independent of the action
  - Proof: repeat calculation

Same expectation – less variance

Challenge of Variance III

- Good pass!
- But striker missed $\Rightarrow \quad R = 0$



Challenge of Variance III

- Problem: Return of a_t depends on future…
  - actions $a_{t+1}, a_{t+2}, \ldots$ (striker)
  - transitions (wind, goalkeeper)

Challenge of Variance III

- Problem: Return of a_t depends on future…
  - actions $a_{t+1}, a_{t+2}, \ldots$ (striker)
  - transitions (wind, goalkeeper)

- Solution:
  - Instead of: $R(\tau) - V(s)$
  - Use ***advantage***: $A \coloneqq r + V(s') - V(s)$

Goal / no goal?

Increased potential for goal

Challenge of Variance III

- Problem: Return of a_t depends on future…
  - actions $a_{t+1}, a_{t+2}, \dots$ (striker)
  - transitions (wind, goalkeeper)

- Solution:
  - Instead of: $R(\tau) - V(s)$
  - Use ***advantage***: $A \coloneqq r + V(s') - V(s)$

Goal / no goal?

Increased potential for goal

<u>Tradeoff</u>

aleatoric noise (in r)

epistemic noise (in V)

Challenge of Variance III

- Problem: Return of a_t depends on future…
  - actions $a_{t+1}, a_{t+2}, \dots$ (striker)
  - transitions (wind, goalkeeper)

- Solution:
  - Instead of: $R(\tau) - V(s)$
  - Use ***advantage***: $A \coloneqq r + V(s') - V(s)$

Goal / no goal?

Increased potential for goal

Tradeoff

aleatoric noise (in r) $\sim TD(1)$

epistemic noise (in V) $\sim TD(0)$

Challenge of Variance III

- Problem: Return of a_t depends on future…
  - actions $a_{t+1}, a_{t+2}, \dots$ (striker)
  - transitions (wind, goalkeeper)

- Solution:
  - Instead of: $R(\tau) - V(s)$ — Goal / no goal?
  - Use ***advantage***: $A \coloneqq r + V(s') - V(s)$ — Increased potential for goal

Tradeoff	
aleatoric noise (in r)	$\sim TD(1)$
epistemic noise (in V)	$\sim TD(0)$

- Control the tradeoff: ***Generalized Advantage Estimation*** (*GAE*) $\sim TD(\lambda)$

Summary

- Deep RL: learn *across* states
  - Approx. planning
  - DQN
  - Policy-Gradient

$$\nabla_\theta E_{\tau\sim\pi_\theta}[R] \approx \frac{1}{N}\sum_{i=1}^{N}\sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_{i,t}|s_{i,t})\, R(\tau_i)$$

    - REINFORCE: G_t
    - Actor-Critic: $G_t - V(s)$
    - Advantage: $r + V(s') - V(s)$